

YUXI HONG

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EDUCATION

King Abdullah University of Science and Technology (KAUST), Thuwal, KSA

August 2019 - June 2023(expected)

Ph.D. in Computer Science, GPA: 3.6/ 4.0, Advisor: [David E. Keyes](#)

Research interest: Numerical Linear Algebra, GPU programming, Machine Learning

Tsinghua University (THU), Beijing, China

August 2015 - June 2018

M.Eng in Microelectronics, GPA: 3.0/4.0, Advisor: [Zuochang Ye](#)

Thesis Topic: Statistical Learning in Semiconductor Modeling

Tsinghua University (THU), Beijing, China

August 2011 - June 2015

B.Eng in Electrical Engineering, GPA: 85/100

RESEARCH PROJECTS

Tile Low-Rank Matrix Vector Multiplication (TLR-MVM)

Advisor: David E. Keyes, [Hatem Ltaief](#), [Matteo Ravasi](#) at KAUST

Matrix-Vector Multiplication (MVM) is a fundamental memory-bound Level-2 BLAS operation. The kernel drives the performance of various scientific applications, including 1) seismic imaging to reveal and monitor the subsurface for a sustainable management of energy resources, and 2) ground-based computational astronomy for supporting real-time simulations necessary to outsmart the atmospheric turbulence and help identifying exoplanets. We further leverage the inherent data sparsity structure of the corresponding matrices using Tile Low-Rank (TLR) matrix approximations. Our TLR-MVM outperforms its dense counterpart on many vendor architectures with high productivity in mind and maintains the numerical robustness of the applications.

Asynchronous Stochastic Gradient Descent

Advisor: [Torsten Hoefer](#), Tal Ben-Nun at ETH

We achieve an optimal tradeoff between allreduce operation's efficiency and convergence during the multi-node training purpose of a deep neural network. FFLib2, a fully asynchronous library is used for this purpose. LibLSB in SPCL is also utilized to establish performance modeling.

Multivariate Rational Regression in Semiconductor Device Modeling

Advisor: Zuochang Ye at THU

We use multivariate rational regression to fit the current voltage characteristic curve of new semiconductor devices for fast modeling. The new model gives better performance than traditional regression models.

PUBLICATIONS

1. **Yuxi Hong**, Hatem Ltaief, Matteo Ravasi, Laurent Gatlaveau, and David Keyes. 2022. High-Performance Seismic Redatuming by Inversion Using Algebraic Compression and Mixed Precisions. (Submitted to ACM TOMS)
2. Matteo Ravasi, **Yuxi Hong**, Hatem Ltaief, David Keyes. 2022. Tile-Low Rank Compressed Multi-Dimensional Convolution and Its Application to Seismic Redatuming Problems. Accepted by 83rd [EAGE](#) Annual Conference & Exhibition. [DOI](#)

3. Matteo Ravasi, **Yuxi Hong**, Hatem Ltaief, David Keyes, David Vargas. 2022. Large-scale Marchenko imaging with distance-aware matrix reordering, tile low-rank compression, and mixed-precision computations. Accepted by **IMAGE**, Second International Meeting for Applied Geoscience & Energy. [DOI](#)
4. **Yuxi Hong**, Hatem Ltaief, Matteo Ravasi, Laurent Gatlneau, and David Keyes. 2021. Accelerating Seismic Redatuming Using Tile Low-Rank Approximations on NEC SX-Aurora TSUBASA. Accepted by **Supercomputing Frontiers and Innovations**, 8(2), 6-26. [DOI](#)
5. Hatem Ltaief, Jesse Cranney, Damien Gratadour, **Yuxi Hong**, Laurent Gatlneau, and David Keyes. 2021. Meeting the real-time challenges of ground-based telescopes using low-rank matrix computations. In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis **SC 21**. Association for Computing Machinery, New York, NY, USA, Article 29, 1-16. [DOI](#)
6. **Yuxi Hong**, El Houcine Bergou, Nicolas Doucet, Hao Zhang, Jesse Cranney, Hatem Ltaief, Damien Gratadour, Francois Rigaut, and David Keyes. 2021. Outsmarting the Atmospheric Turbulence for Ground-Based Telescopes Using the Stochastic Levenberg-Marquardt Method. In Proceedings of **EuroPar 21**, 27th International Conference on Parallel and Distributed Computing, Lisbon, Portugal, September 1-3, 2021. [DOI](#)
7. Hao Zhang, Jesse Cranney, Nicolas Doucet, **Yuxi Hong**, Hatem Ltaief, Damien Gratadour, Francois Rigaut, David Keyes. Predictive learn and apply: MAVIS application - learn. Accepted by **Adaptive Optics Systems VII 2020** [DOI](#)
8. Jesse Cranney, Nicolas Doucet, Hao Zhang, **Yuxi Hong**, Hatem Ltaief, Damien Gratadour, Francois Rigaut, David Keyes. Predictive learn and apply: MAVIS application - apply. Accepted by **Adaptive Optics Systems VII 2020** [DOI](#)
9. **Yuxi Hong**, Peng Han. LSDDL: Layer-wise Sparsification for Distributed Deep Learning. Accepted by **Big Data Research**, Volume 26, Page 100272. [DOI](#)
10. **Yuxi Hong**, Ling Chen, & Zuochang Ye (2018, April). End-to-end soccer video scene and event classification with deep transfer learning. Accepted by 2018 International Conference on Intelligent Systems and Computer Vision, **ISCV** (pp. 1-4). IEEE. [DOI](#)
11. **Yuxi Hong**, Dongshen Ma, & Zuochang Ye (2018). Multivariate rational regression and its application in semiconductor device modeling. Accepted by **Journal of Semiconductors**, 39(9), 094010. [DOI](#)

COMMUNITY SERVICES

1. Reviewer of academic conferences
 - IEEE Cluster Conference 2019 [\[Link\]](#)
 - The Platform for Advanced Scientific Computing Conference (PASC) 2020 [\[Link\]](#)
 - ISC High Performance 2020 [\[Link\]](#)
2. President of Student Chapter of Association for Computing Machinery (ACM) at KAUST

HONORS & AWARDS

- KAUST Ph.D. Fellowship
- 2nd Place in KAUST Winter Enrichment Program Personalized Medicine Mobile App Development
- Tsinghua-NUS-IBS Asian Entrepreneurship and Innovation summer program (AIEIP)
- Attend Open Geneva Hackathon as a representative of Tsinghua University (about \$5000)

- Tsinghua-UCSD Future Global Leadership Program (about \$5000)